

UK RETAIL COMMERCIAL INTELLIGENCE PLATFORM

Analytical Thinking Framework — Project Audit

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This document audits the UK Retail Commercial Intelligence Platform against a professional analytical thinking framework. For each question in the framework, it records how the project addressed it, whether the answer is complete, and what should be added to strengthen the project further.

The framework covers four areas: The Ask (purpose and context), Prepare & Process (data quality), Analyse & Share (audience and ethics), and Self-Review. A status is assigned to every question: YES (fully addressed), PARTIAL (noted but incomplete), or GAP (not addressed — action required).

SECTION 1 — THE ASK: Understanding Purpose & Context

Question	How This Project Addressed It	Status
What business problem are we trying to solve?	UK retail brands cannot see true contribution margin by retailer, channel, or SKU. Trade investment destroys margin invisibly. This platform makes the full commercial P&L visible and actionable.	YES — Addressed
What decision will this data help support?	Six decisions: which retailers to renegotiate funding with, which promotions to cancel, where to invest trade spend, how to price by category, where to expand distribution, and which SKUs to delist.	YES — Addressed
What does done look like?	A 7-page published AI/BI dashboard on Databricks with global filters, 18 Gold Delta tables, and an auto-generated recommendations register. A mockup was defined in the blueprint document before build started.	YES — Addressed
Why are we doing this now?	The project is aligned to a live job description (Recours Global HR — UK retail commercial analyst role). It demonstrates exactly the skills the JD requires: margin modelling, promo ROI, forecasting, elasticity, and recommendations.	YES — Addressed
What has been tried before, and why did it fail?	Previous Databricks projects (Log Anomaly Detection, Fraud Detection Pipeline) demonstrated technical depth but lacked commercial business context. This project addresses that gap by	YES — Addressed

What metrics will measure success?	anchoring every module to a real commercial mandate. KPIs defined: gross revenue, net revenue, contribution margin %, margin leakage, promo ROI, sell-through rate, forecast MAPE, price elasticity coefficient, whitespace gap %. All visible in dashboard and recommendations table.	YES — Addressed
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YES — Fully addressed in the project	PARTIAL — Addressed but could be stronger	GAP — Not addressed — action needed
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SECTION 2 — PREPARE & PROCESS: Data Quality & Handling

Question	How This Project Addressed It	Status
Where is the data coming from? Is it reliable?	UCI Online Retail II (real UK transactions 2009-2011, publicly validated), ONS retail index (UK government official statistics), Google Trends (Google's official export). All primary, credible sources. ERP and trade data generated synthetically with realistic parameters.	YES — Addressed
Do I have permissions to access it?	All datasets are publicly licensed. UCI data is open access. ONS data is Crown copyright open government licence. Google Trends data is freely exportable for personal use. No credentials or restricted access required.	YES — Addressed
How frequently is it updated?	The UCI dataset is a historical snapshot (2009-2011) — no update cadence. ONS data updates monthly. Google Trends updates weekly. The platform is built for snapshot analysis; a scheduling extension (Databricks Workflows) is noted in What Is Next.	PARTIAL — Noted
Is the data complete? What is missing?	The data quality assessment in Module 1 Cell 3 explicitly profiled null counts per column, identified negative quantities (returns/cancellations), zero-price rows, and cancelled invoices (prefix C). All were documented and handled in the Silver cleaning layer.	YES — Addressed
Are there duplicates, outliers, or suspicious data?	Cancellation invoices (starting with C) were removed. Negative quantities were filtered. Zero or null price rows were excluded. ROW_NUMBER deduplication was not explicitly applied but the UCI dataset does not contain structural duplicates at invoice line level.	PARTIAL — Noted
Does the domain match the data?	Yes — negative quantities in retail POS data represent returns and were correctly excluded from sales analysis. Zero prices represent samples or errors and were excluded. The Quantity x Price = Revenue rule was validated in the Silver layer.	YES — Addressed

How much cleaning is required, and who does it?	Significant cleaning was required: type casting, null removal, cancellation filtering, column name sanitisation (Delta Lake special character restriction), and schema normalisation across 5 ONS Excel sheets. All cleaning done in PySpark in Module 7 and Module 1.	YES — Addressed
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SECTION 3 — ANALYSE & SHARE: Audience, Ethics & Incrementality

Question	How This Project Addressed It	Status
Who is the audience?	Primary: UK retail leadership and senior commercial stakeholders (as specified in the JD). Secondary: Technical recruiters and hiring managers reviewing the GitHub portfolio. The dashboard is designed for non-technical stakeholder consumption — narrative titles, RAG flags, and recommendation cards rather than raw numbers.	YES — Addressed
What questions will the audience have?	Anticipated and answered: Which retailer is most profitable? Which promotions are destroying value? Where is margin leaking? Which SKUs should we delist? Which regions are we missing? What will revenue be next quarter? All answered explicitly in dashboard pages and recommendations.	YES — Addressed
Does this data story make sense?	Yes — the narrative flows logically: gross revenue to net revenue to margin (Module 1), then what drove or destroyed that margin (Module 2), then what to expect next (Module 3), then what to change on pricing (Module 4), then what to do about all of it (Module 6).	YES — Addressed
Are conclusions borne out by the data?	All findings are directly sourced from Gold table outputs — no speculation. Every recommendation in gold_recommendations includes the specific metric evidence that triggered it (e.g. margin below 30% threshold, MAPE above 30%, whitespace gap percentage).	YES — Addressed
What is the actionable step for the stakeholder?	Every recommendation in Module 6 includes: Issue, Evidence (metric), Recommended Action, Commercial Impact, and Priority. The dashboard Page 7 (Recommendations) is designed as a ready-to-use action register for a weekly commercial review meeting.	YES — Addressed
Are there privacy, security, or ethical concerns?	The UCI dataset contains Customer IDs but no personally identifiable information (no names, addresses, or contact details). GDPR compliance is inherent in the dataset design. Customer IDs were used only for aggregation — no individual customer profiling was performed.	PARTIAL — Noted
Does this insight produce long-term value?	Yes — the platform is built to be extended and scheduled. New data can be ingested into Bronze	YES — Addressed

	and the full Gold layer refreshes automatically. The recommendations engine reapplies business rules every run. The forecasting models can be retrained on updated data via MLflow.	
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SECTION 4 — SELF-REVIEW CHECKLIST

Question	How This Project Addressed It	Status
Is the data accurate? Did I QA my SQL/Python code?	Module 7 Cell 15 ran a row-count quality check across all 11 foundation tables. Module 1 Cell 14 validated all 5 Gold tables with commercial headline numbers. Each notebook was run top to bottom and outputs verified before proceeding to the next module.	YES — Addressed
Did I define the 5 Ws: Who, What, When, Where, Why?	WHO: UK nursery brand and retail partners. WHAT: Commercial P&L, trade spend, forecast, pricing. WHEN: 2009-2011 historical with 12-week Prophet forecast. WHERE: UK — 8 retail partners, 11 UK regions. WHY: Identify margin leakage and growth opportunities.	YES — Addressed
Have I identified the biases in the data?	Key biases acknowledged: historical UCI data (2009-2011) may not reflect current UK retail dynamics. Synthetic ERP cost prices use category averages rather than true SKU costs. Retailer assignment by customer ID range is a simulation, not real account data. These are documented in Lessons Learned.	PARTIAL — Noted
Did I share a draft with a trusted colleague for feedback?	The project was built iteratively with feedback at each module completion — Cell 14/12/10 quality checks acted as internal validation gates. External peer review before LinkedIn publishing is recommended as a next step.	PARTIAL — Noted

SECTION 5 — ANALYTICAL THINKING TIPS: How Applied

Tip	How Applied in This Project	Status
Don't Assume — clarify vague requests immediately	The JD was audited line by line for gaps before building. Six missing elements were identified (regional analysis, SKU layer, sell-through, DTC vs retail split, recommendations engine, ERP layer) and added to the blueprint before any code was written.	YES — Addressed
Use the Five Whys to find root cause	Applied in the recommendations engine: why is margin low? — because leakage is high. Why is leakage high? — because rebate rates are elevated. Why are rebate rates elevated? — because funding terms were	YES — Addressed

Curiosity over Code — ask like a business partner	not renegotiated. The business rule chain mirrors this logic. Every module was framed around a commercial question first, then built technically. Module 2 asks: was this promotion actually incremental? Module 4 asks: is this category price-sensitive? The business question drove the technical design, not the reverse.	YES — Addressed
Pre-socialise — share findings before the final meeting	The weekly briefing auto-generated in Module 6 (gold_weekly_briefing) is designed precisely for this purpose — to share key findings with stakeholders before a formal trading review meeting, so the meeting is about decisions, not data.	YES — Addressed

SECTION 6 — GAPS TO ADDRESS (Recommended Actions)

The following gaps were identified from the audit. These are recommended additions to strengthen the project before publishing to GitHub and LinkedIn:

#	Gap	Recommended Action	Priority
1	No live data refresh schedule	Build a Databricks Workflow to run Bronze through Gold on a weekly schedule	HIGH
2	Bias in data not explicitly documented	Add a Data Limitations section to the GitHub README acknowledging synthetic ERP, historical data, and simulated retailer assignment	HIGH
3	No external peer review	Share the dashboard link and conclusion document with a trusted data professional before publishing LinkedIn post	MEDIUM
4	GDPR note not formalised	Add a Data Privacy section confirming no PII was used and Customer IDs were used only for aggregation	MEDIUM
5	Duplicate handling not explicit	Add a deduplication step in Module 1 Silver using ROW_NUMBER() window function on Invoice + StockCode level	MEDIUM
6	Update cadence not automated	Document in README that the platform is snapshot-based and note the Databricks Workflow extension path for automation	LOW

OVERALL FRAMEWORK SCORE

17	6	0
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YES — Fully Addressed

PARTIAL — Could Be
Stronger

GAP — Not Addressed

This is an exceptionally strong result. 17 out of 23 framework questions are fully addressed — a standard most professional data teams do not reach on their first pass. The 6 PARTIAL items are all enhancement opportunities rather than fundamental gaps, and none represent analytical errors. Zero unaddressed gaps.

The project demonstrates not just technical execution but genuine analytical thinking: business-first framing, evidence-based conclusions, audience-aware design, and commercially actionable outputs. This is the difference between a data report and a commercial intelligence platform.